

Endoscopic resection and transoral laser microsurgery (TLM)

If you have early-stage laryngeal cancer, your surgeon may be able to remove the cancer through your mouth. Sometimes they use a laser to do this.

On this page

- [What is an endoscopic resection and transoral laser microsurgery \(TLM\)?](#)
- [Before an endoscopic resection and transoral laser microsurgery \(TLM\)](#)
- [Having an endoscopic resection and transoral laser microsurgery \(TLM\)](#)
- [After an endoscopic resection and transoral laser microsurgery \(TLM\)](#)
- [About our information](#)
- [How we can help](#)

What is an endoscopic resection and transoral laser microsurgery (TLM)?

If you have early-stage [laryngeal \(larynx\) cancer](#), your surgeon may be able to remove the cancer through your mouth. This is called transoral resection.

You will not have a wound in your neck afterwards. You have it done under a general anaesthetic, and you only need a short stay in hospital.

It may help to reduce the risks of certain side effects, such as swallowing difficulties. It will also help you recover faster.

The main aim of surgery is to remove the cancer completely. Your specialist surgeon will also do everything possible to reduce changes to your speech and swallowing.

Before an endoscopic resection and transoral laser microsurgery (TLM)

Before you have the operation, your surgeon and nurse will explain what will happen before, during and after the surgery.

It is important you understand what the operation involves and how it will affect you in the short and long term.

You will usually meet other members of the [multidisciplinary team \(MDT\)](#) before your surgery. This can include a specialist nurse, a speech and language therapist (SLT) and a dietitian. They can provide support and advice before and after the operation.

You will have some tests before your operation to check:

- how well you would cope with an anaesthetic, if you need one
- whether you can make a good recovery after surgery.

You may also have:

- [blood tests](#)
- a chest [x-ray](#)
- an [electrocardiogram \(ECG\) to check your heart](#)
- breathing tests (lung function tests).

You may have these done at a pre-assessment clinic before you go into hospital.

Having an endoscopic resection and transoral laser microsurgery (TLM)

The surgeon passes a thin, flexible tube with a camera at the end into the mouth and down the throat. This is called an endoscope.

They use the camera to see images of [the larynx](#). These appear on a screen in the operating theatre. This lets your surgeon see the cancer clearly.

They then guide small surgical instruments through the endoscope to remove the cancer.

Types of operation

Transoral laser microsurgery (TLM)

Sometimes, your surgeon uses a high-powered laser to remove the cancer. This is called transoral laser microsurgery (TLM).

The surgeon points the laser beam at the tumour to remove it. They also remove a small amount of healthy tissue around it, called a margin. This is to try to make sure they have removed all the cancer cells. The laser can be used to stop any bleeding during surgery.

Transoral robotic surgery (TORS)

Sometimes, instead of holding the surgical instruments themselves, the surgeon uses a machine or robot to hold them. The surgeon controls the robotic arms. These arms can move very steadily and precisely to remove the tumour through the mouth. This is called transoral robotic surgery.

(TORS).

TORS is not widely available in the UK. Your cancer doctor can explain if it is suitable for you.

After an endoscopic resection and transoral laser microsurgery (TLM)

You may have some pain in your throat and difficulty swallowing for about 2 days afterwards. Your doctor or nurse can tell you what painkillers to take until this gets better.

If the surgery involved a vocal cord, your voice may be hoarse. Your doctor might ask you to rest it for a few days. For some people, the surgery may cause a permanent change to their voice. If this happens, your speech and language therapist (SLT) can give you advice and support. They may also suggest [voice therapy](#).

Rarely, side effects of a transoral resection include:

- bleeding
- infection
- damage to your teeth.

Your cancer doctor or specialist nurse will give you more information about this before you go home.

We have more information about [recovering from surgery](#).

About our information

References



Below is a sample of the sources used in our laryngeal cancer information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

ESMO Annals of Oncology. Squamous cell carcinoma of the oral cavity, larynx, oropharynx and hypopharynx. EHNS-ESMO-ESTRO Clinical Practice Guidelines for diagnosis, treatment and follow up. 2020. Available from <https://www.esmo.org/guidelines/ehns-esmo-estro-clinical-practice-guideline-squamous-cell-carcinoma-of-the-head-and-neck> (accessed Jan 2022).

NICE Guideline NG36. Cancer of the upper aerodigestive tract: assessment and management in people aged 16 and over. 2018. Available from www.nice.org.uk/guidance/ng36 (accessed Jan 2022).

NICE Technology Appraisal TA736. Nivolumab for treating recurrent of metastatic squamous cell carcinoma of the head and neck after platinum-based chemotherapy. 2021. Available from www.nice.org.uk/guidance/ta736 (accessed Jan 2022).

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, [Dr Claire Paterson](#), Consultant Clinical Oncologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

Date reviewed

Reviewed: 01 June 2022 | Next review: 01 June 2025

⚠ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help